MAE 270C Final Project Presentation

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Problem Formulation: Thrust-Vectored Lander

states: $x = [p_x, p_z, v_x, v_z, \theta, \omega, m]^\top$

control: $u = [\delta T, \tau]^\top, \delta T = T - T*$

nonlinear dynamics:

$\dot{\theta} = \omega, \dot{\omega} = \frac{\tau}{I}, \dot{m} = -\alpha T$

$\dot{p}_x = v_x, \dot{p}_z = v_z, \dot{v}_x = \frac{T}{m} \sin\theta, \dot{v}_z = \frac{T}{m} \cos\theta - g$

Part I: Fixed-Time LQR Around Different Nominal Trajectories

Linearization

$$\dot{x} = Ax + Bu$$

Hover trim (LTI)

v=0, θ=0, T*=mg

$$A_{hover}=\begin{bmatrix} 0&0&1&0&0&0&0\\ 0&0&0&1&0&0&0\\ 0&0&0&0&g&0&0\\ 0&0&0&0&0&0&-\frac{g}{m_0}\\ 0&0&0&0&0&1&0\\ 0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0 \end{bmatrix} \quad B_{hover}=\begin{bmatrix} 0&0\\ 0&0\\ 0&0\\ \frac{1}{m_0}&0\\ 0&0\\ 0&\frac{1}{I}\\ -\alpha&0 \end{bmatrix}$$

Descent Trim (LTV)

`v_x* ≫ 0`, `v_z* < 0`, `T*≈mg`

$$A(t)=\begin{bmatrix} 0&0&1&0&0&0&0\\ 0&0&0&1&0&0&0\\ 0&0&0&0&g&0&0\\ 0&0&0&0&0&0&-\frac{g}{m^*(t)}\\ 0&0&0&0&0&1&0\\ 0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0 \end{bmatrix} \quad B(t)=\begin{bmatrix} 0&0\\ 0&0\\ 0&0\\ \frac{1}{m^*(t)}&0\\ 0&0\\ 0&\frac{1}{I}\\ -\alpha&0 \end{bmatrix}$$

Mass depletion
m˙=−αTm˙=−αT → on trim T*≈mg
T*≈mg → m˙=−αgm
m˙=−αgm → m(t)=m0e−αgtm(t)=m0e−αgt.

Controllability Analysis

Controllability matrix: determine if a dynamic system's internal states can be fully manipulated using its external inputs

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & A^3B & A^4B & A^5B & A^6B \end{bmatrix}$$

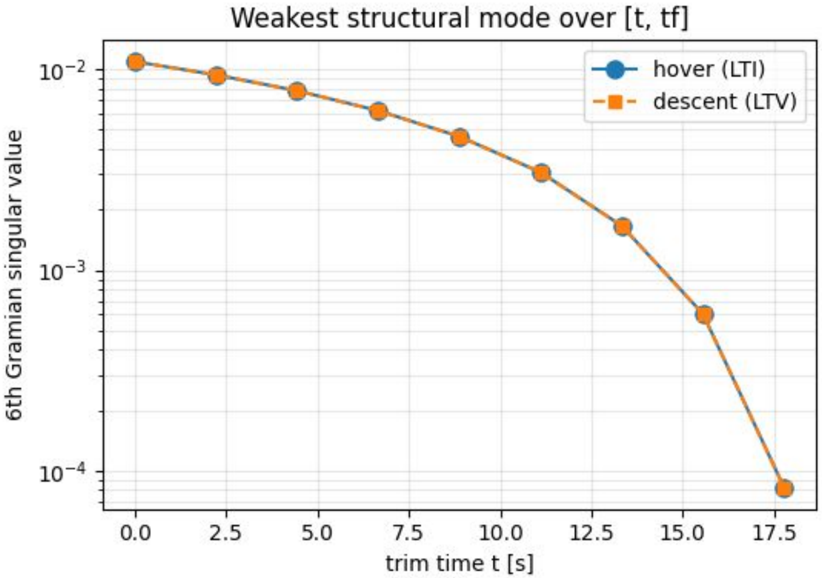
Controllability gramian (finite horizon): accumulates how control can steer the state through the coupled dynamics over time

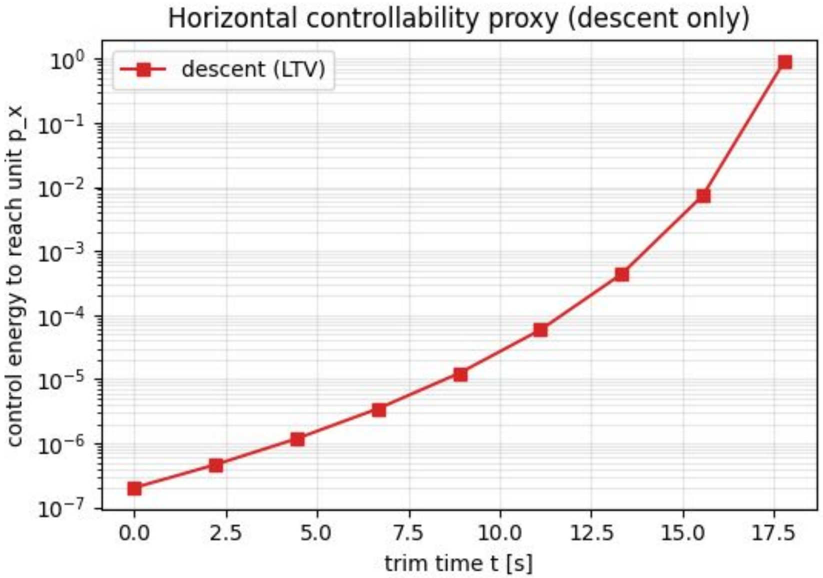
$$W_c(0, t_f) = \int_0^{t_f} \Phi(0, \tau) B(\tau) B(\tau)^\top \Phi(0, \tau)^\top d\tau$$

Gramian simplification to avoid state transition matrix computation

$$\dot{W}_c(t) = A(t) W_c(t) + W_c(t) A(t)^\top + B(t) B(t)^\top, \quad W_c(0) = \mathbf{0}_{7\times 7}$$

Hover has no controllability problem worth investigating, especially horizontal speed





6th mode: weakest structural mode (attitude / horizontal coupling)

- Hover and descent gramian magnitude match on this mode, $\tau \to \omega \to \theta \tau \to \omega \to \theta$. No mass dependence

7th mode: mass depletion (can only be burned not added to)

- Probably should plot 7th node to show discrepancy btwn hover and descent

Global weakest mode (attitude) and horizontal authority both deteriorate with time left.
Descent is where that matters for the mission (landing clock + horizontal corrections).

The horizontal proxy measures how much open-loop control energy (mainly through torque → tilt → $T\sin\theta \approx T\theta$ → vx → px) is needed to affect 1 m of px over the time left until tf (not direct thrust along x, but thrust effectiveness after attitude changes, which gets much worse when little time remains.)

Analytical formulation (PMP, Riccati RDE)

Optimal control problem

$$\boxed{J_{\mathrm{reg}} = \int_0^{t_f} (x^\top Q x + u^\top R u)\, dt + x(t_f)^\top Q_f\, x(t_f)}$$ (running state & control penalty, and terminal penalty)

$$\mathrm{s.t.}\ \dot{x} = A(t)x + B(t)u,\ x(0) = x_0\ \mathrm{given},\ x(t_f)\ \mathrm{free}.$$

$Q \succeq 0, R \succ 0, Q_f \succeq 0$ (diagonal in your numerics).

Step 1 — Hamiltonian (PMP)}

Introduce the costate $\lambda(t) \in \mathbb{R}^n$ and define $H w.r.t.u$

$$\boxed{H(x,u,\lambda,t) = x^\top Q x + u^\top R u + \lambda^\top (A(t)x + B(t)u).}$$

Interpretation: optimal control balances control penalty $2Ru$ against the influence of the costate on the plant through $B^\top \lambda$

Step 2 - Stationarity Conditions

$$\frac{\partial H}{\partial u} = 2R\,u + B(t)^\top \lambda = 0 \implies \boxed{u^*(t) = [\backslash\mathrm{tfrac?}]12\, R^{-1} B(t)^\top \lambda(t).}$$

Analytical formulation (PMP, Riccati RDE)

Step 3 — State equation

$$\boxed{\dot{x} = A(t)x + B(t)u^*,\ x(0) = x_0.}$$

Step 4 — Adjoint (costate) equation}

Along an optimal arc,

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = -\left(2Q\,x + A(t)^\top \lambda\right),$$

equivalently

$$\boxed{-\dot{\lambda} = 2Q\,x + A(t)^\top \lambda.}$$

Analytical formulation (PMP, Riccati RDE)

Step 5 — Transversality (free endpoint)}

Because **x(t_f)** is free and the terminal payoff is , $\Phi(x_f) = x_f^\top Q_f x_f$

Physical meaning: if the terminal state is still large, the costate at t_f is large, pushing the backward integration to aggressively correct the trajectory.

$$\boxed{\lambda(t_f)=\frac{\partial \Phi}{\partial x}[\backslash\mathrm{Big}|?]_{x(t_f)}=2Q_f\,x(t_f).}$$

Step 6 — Closed-loop TPBVP

Substitute u^* into xdot, lambdadot.

$$\begin{cases} \dot{x}=Ax-[\backslash\mathrm{tfrac?}]12BR^{-1}B^\top\lambda, \\ -\dot{\lambda}=2Qx+A^\top\lambda, \\ \lambda(t_f)=2Q_f x(t_f). \end{cases}$$

This two-point boundary-value problem (TPBVP) is what LQR reduces analytically.

Analytical formulation (PMP, Riccati RDE)

Step 7 — Riccati reduction (quadratic ansatz)
Assume a linear costate map (state feedback structure): $\boxed{\lambda(t) = P(t)\,x(t).}$

Then $u^*(t) = [\backslash\mathrm{tfrac?}]12\, R^{-1}B^\top P(t)\,x(t) \equiv -K(t)\,x(t),\ K(t) = [\backslash\mathrm{tfrac?}]12\, R^{-1}B^\top P.$

Substitute lambda=Px into the adjoint and match the x-coefficient (TPBVP -> matrix ODE):

$$\boxed{-\dot{P} = A^\top P + PA - [\backslash\mathrm{tfrac?}]12\, PBR^{-1}B^\top P + 2Q,\ P(t_f) = Q_f.}$$

This is the **matrix Riccati differential equation (RDE)** integrated **backward** from **t_f** to **0**.

$$\boxed{u^*(t) = K(t)\,x(t),\ K(t) = [\backslash\mathrm{tfrac?}]12\, R^{-1}B(t)^\top P(t).}$$

Optimal regulation law (feedback form).

$$-\dot{P} = A^\top P + PA - PBR^{-1}B^\top P + Q,\ P(t_f) = Q_f,\ u^* = R^{-1}B^\top Px,$$

Solver

2 pass sweep

- BACKWARD ($t_f \rightarrow 0$): integrate RDE $\rightarrow$ $P(t)$ on grid

$K(t) = R^{-1} B(t)^T P(t)$ $\leftarrow$ gain is computed from P

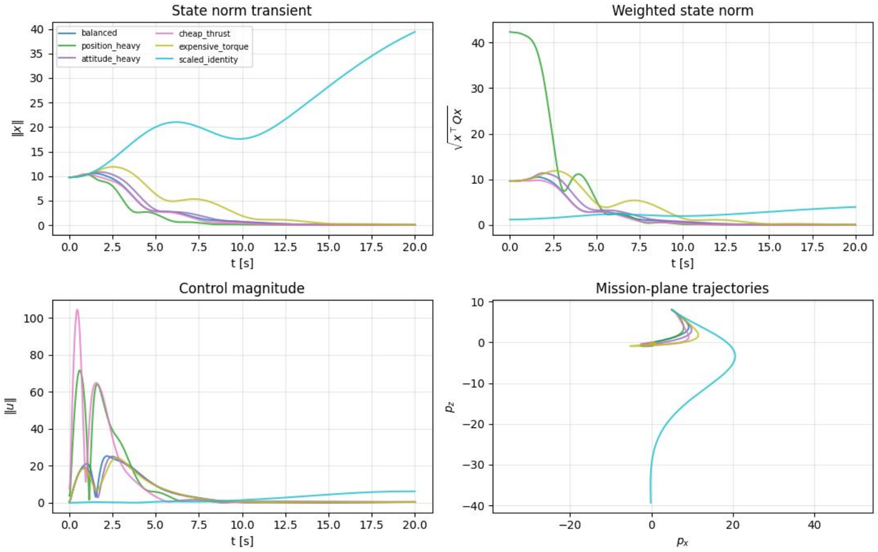
- FORWARD ($0 \rightarrow t_f$): simulate $\dot{x} = A(t)x + B(t)u^*(t)$ with $u^* = -K(t)x$ $\rightarrow$ $x(t)$, $u(t)$
- EVALUATE: J, $||x(tf)||$, transients, IC sweeps

Numerical Analysis

Q, R,Q_f comparisons (hover LTI)

Perturbation transients, deviation from trim

shows transient in the same units as the objective; highlights attitude-heavy presets and where descent pays extra ‘cost-weighted’ effort even when // x // looks similar.



Trajectory quality heuristics

(for picking weights):

- terminal state norm small $\|x(t_f)\|$

- no excessive control peaks or saturation risk

- smooth decay of $\sqrt{x^\top Qx}$

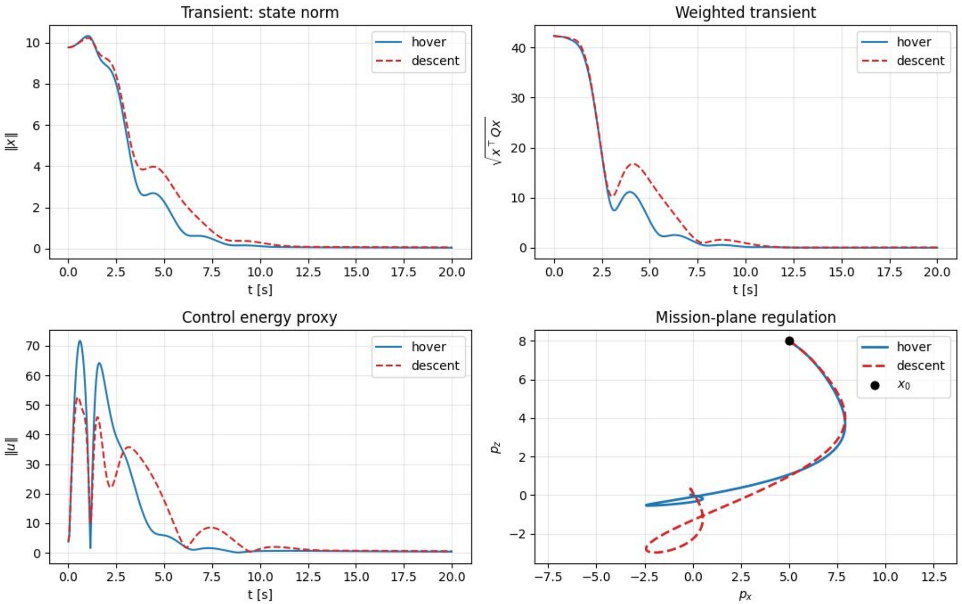
(no large overshoot in attitude/position)

Normalizing reference:

- DEFAULT_REFERENCE_SCALES = np.array([10.0, 10.0, 5.0, 5.0, 0.2, 0.5, 1.0])
- Interpretation: “a typical perturbation” is ~10 m position, ~5 m/s velocity, ~0.2 rad attitude, etc.

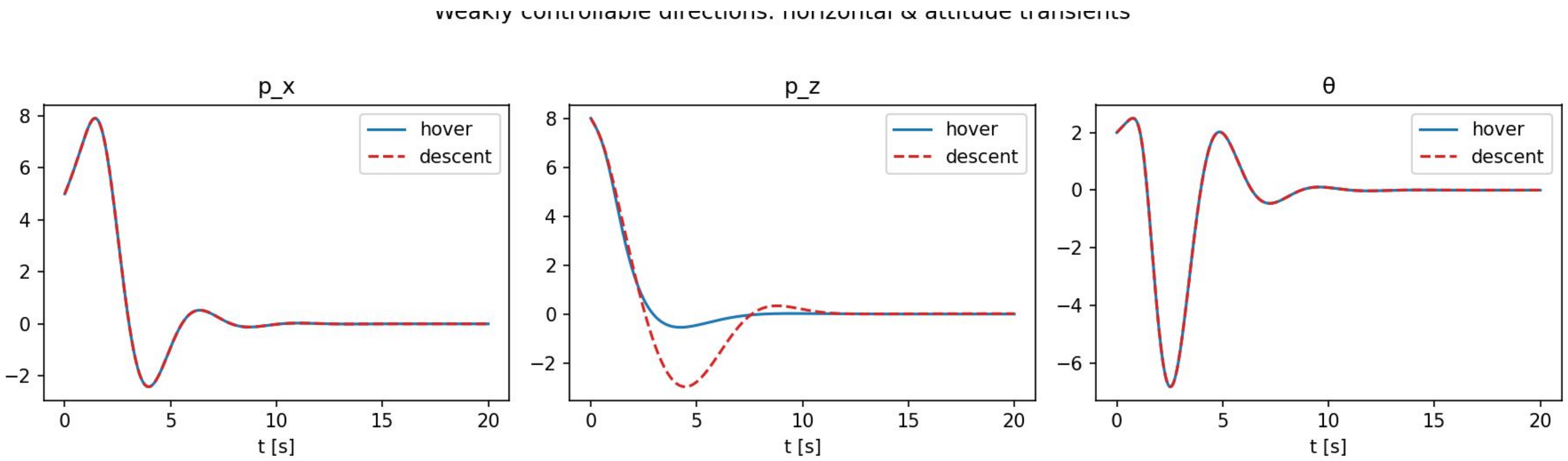
Controllable vs weakly controllable linearizations

Descent is worse on cost and terminal accuracy, but not because it burns more control. Uses a different K(t) on a harder LTV plant. The penalty shows up as slower decay, a mid-horizon rebound, and a messier path, not necessarily higher peak thrust.



Controllable vs weakly controllable linearizations

1. px and θ swings = structural difficulty. Even “controllable” hover struggles. Gramian analysis flagged as expensive to correct.
2. The gap is clearest in pz (mass/thrust schedule, changing trim). Descent worse



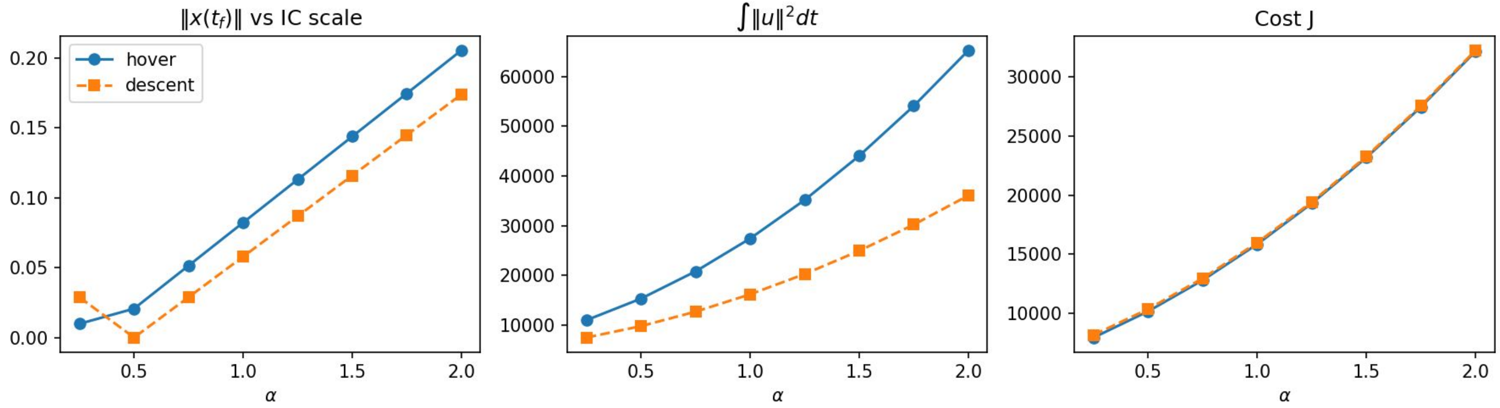
Controllable vs weakly controllable linearizations

IC sensitivity: sweep across multiple initial perturbations

End result

Actuation price only

Full objective score



- At small perturbations: descent can be worse (weak plant + wrong linearization point).
- At large perturbations: hover overtakes — hover LTI may be overly optimistic at large α (linear model stays fixed while state explores larger deviations).

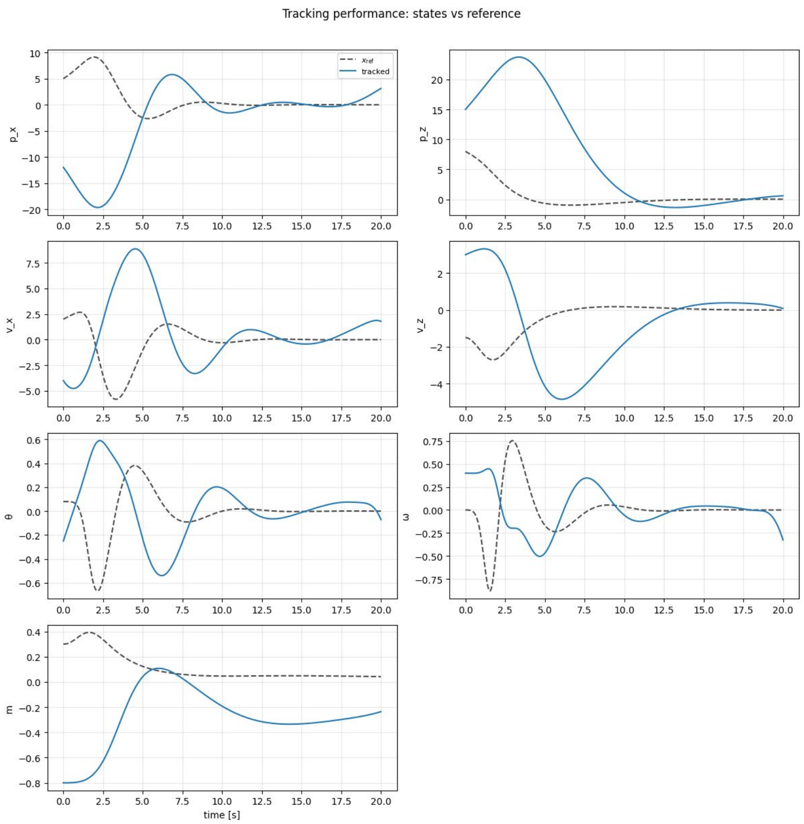
Part II: Optimal Trajectory Tracking

Trajectory tracking

finite-horizon tracking problem $$\min_u \int_0^{t_f} \left[(x - x_{\mathrm{ref}})^\top Q(x - x_{\mathrm{ref}}) + u^\top Ru\right] dt.$$

- Finite-horizon trajectory tracking: over [0,tf], minimize running cost on deviation from a pre-planned reference, subject to linearized lander dynamics, with free terminal state and no terminal cost.
- Given a known reference path, find the LQR controller that minimizes weighted tracking error plus control effort over a fixed horizon; simulate from a badly misplaced initial state and see if the vehicle can catch the plan.
- Tested in hover LTI

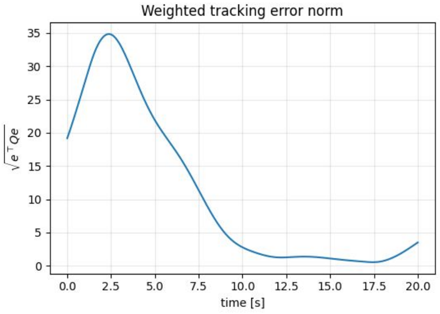
Tracking Performance

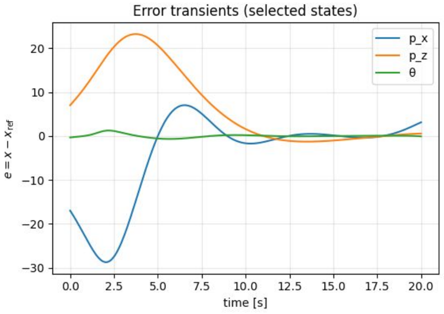


Tracker rejoins the reference qualitatively

errors shrink and shapes align by mid-horizon.

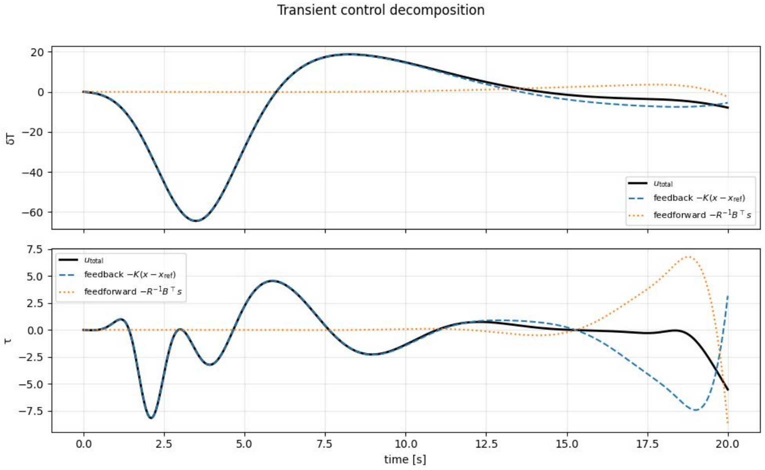
Transient Response





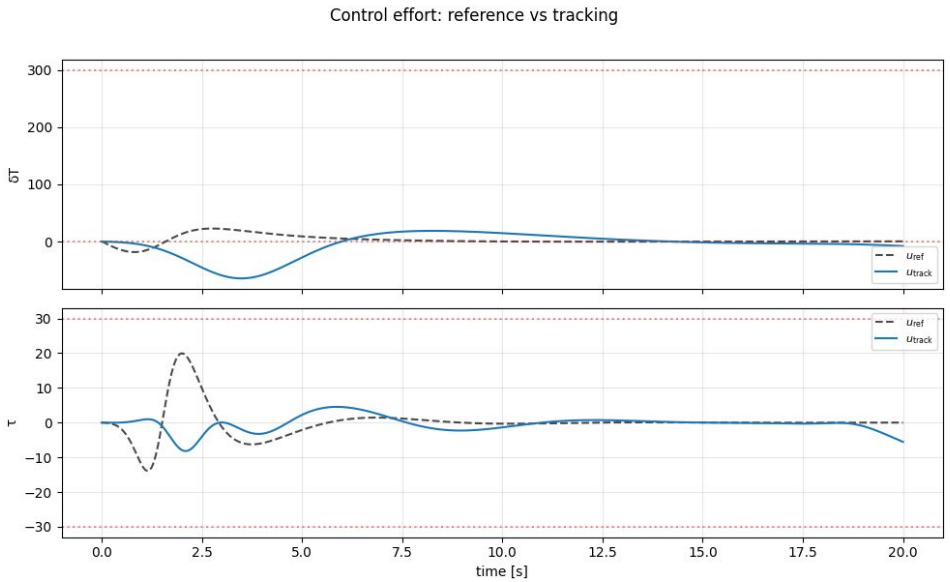
scalar tracking error (cost-balanced error) decays monotonically in the weighted norm. Moving target, not zero, so no rebound mid-horizon

Position errors decay slowly and dominate; attitude error stays small throughout. Playing ‘catch-up’



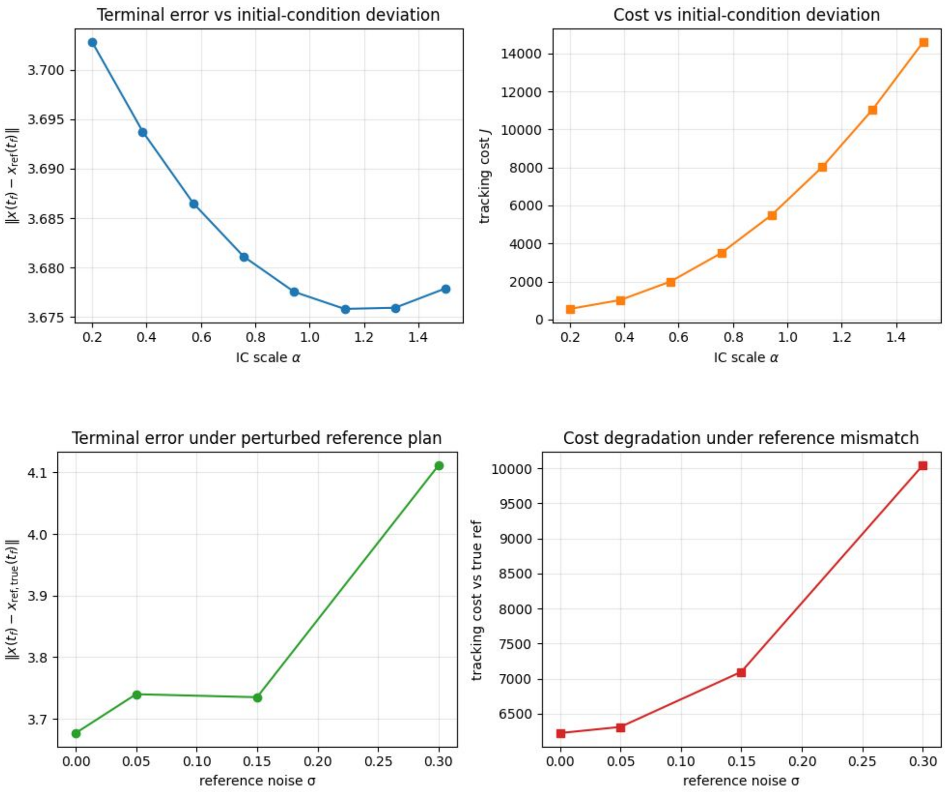
Feedback dominant early, feedforward term reduces lag later

Control Effort



- Reference control: cost of generating the nominal mission
- Cost of rejoining that mission from a badly misplaced start
- Reference control comes before tracking control

Robustness to deviations from the nominal trajectory



Part III: The Mission Problem

Part IV: The Nonlinear Optimal Control Near Hover